Name: _____ Class and student number: _____

Check answers

Read this booklet only after completing and keeping first answers in B. Use a different pen colour to check and correct; do not erase first answers.

The trace is higher in the later image: a clue to upward transport, not a direct view of water vapour (水汽) lost.

No: “better” has no defined measure. The factor is leaf area; replace “better” with faster average upward speed (cm/min), keeping other conditions the same. Testable (可測試的) does not mean proved.

① Petiole (葉柄), xylem (木質部); ② 0 cm; ③ 2 cm. Use A/B/C for light, B/D for leaves. A/D changes light and leaf presence together and cannot isolate the leaf factor.

Group	Factors
Independent variable	Light intensity (光強度)
Dependent variable	Height / average upward speed of the red water trace
Controlled variables	Temperature / Humidity (濕度) and air movement / Red water concentration and depth of immersion (浸入深度) / Total leaf area of the setups with leaves / Length and thickness of the celery stalks

What assumptions does this investigation make?

Changing the lamp distance does not make the setups have different temperatures. / Setups A, B and C have the same total leaf area. / Humidity (濕度) and air movement are the same for all setups.

Monitor and maintain 25°C so lamp heating does not become another factor affecting transpiration (蒸騰) and transport. Also control leaf area, water supply, humidity (濕度) and airflow; do not merely assume no heating.

Model readings and mean speeds

min	A (cm)	B (cm)	C (cm)	D (cm)
0	0.0	0.0	0.0	0.0
10	4.0	2.5	1.5	0.5
20	8.0	5.0	3.0	1.0
30	12.0	7.5	4.5	1.5

Setup	Average upward speed (cm/min)
A	0.40
B	0.25
C	0.15
D	0.05

1. How is light intensity (光強度) related to the upward speed of the water trace? increase

2. What is the role of leaves? faster

2. What is the role of leaves? help water move upwards

3. What can this experiment tell us? The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率).

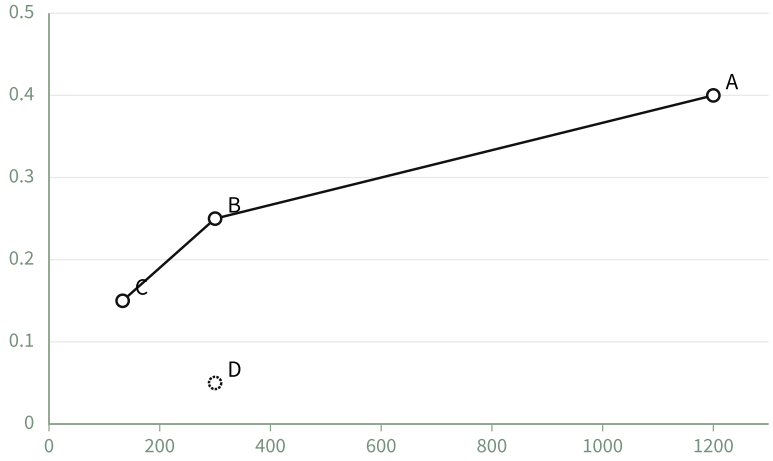
Trace speed is indirect evidence of transport, not a direct transpiration rate (蒸騰速率); water vapour (水汽) loss was not measured. Leaves/no leaves alone cannot prove that water vapour (水汽) escapes only through stomata (氣孔).

Untested light intensities cannot be predicted exactly.

Check the graph, explain and correct

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Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

The model gives 0.15, 0.25 and 0.40 cm/min for the group with leaves at 133, 300 and 1200 lux. At 300 lux, B is 0.25 and D is 0.05 cm/min, supporting a leaf role in upward transport. Three points do not establish an exact untested curve.

Use A/B/C and B/D data to explain light, transpiration (蒸騰) and xylem (木質部) transport, including suitable conditions and evidence limits.

Choose a wrong or incomplete answer of your own: state it and correct it with data or concepts. If none, correct “trace speed is transpiration rate (蒸騰速率)”.

Explanation reference (try your own explanation first):

With suitable conditions and enough water, stronger light usually promotes stomatal opening (氣孔張開) and transpiration (蒸騰). Water evaporates (蒸發) from mesophyll surfaces; vapour diffuses through stomata (氣孔). Water potential falls, water is replaced by osmosis and pulled from xylem (木質部), forming transpiration pull (蒸騰拉力). Data support a transport trend, not a direct transpiration rate (蒸騰速率).

State the original error and correct it using data or concepts. Example: replace “trace speed equals transpiration rate (蒸騰速率)” with “a transport clue; water vapour (水汽) loss needs a separate measurement”. Accept other reasonable corrections.